1 The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ the liquid. The speed is given by the equation

$$v = k\left(\frac{\Delta P}{\rho}\right)^n$$

where k is a constant that has no unit.

What is the value of n?

A 0.5 **B** 1 **C** 1.5 **D** 2

2 The diameter D of a sphere is measured to be 5.0 cm with a fractional uncertainty of 0.02